

Global Concerns about food security and safety

Definitions

Food insecurity

Food insecurity is defined by the The Food and Agriculture Organization (FAO) as “A situation that exists when people lack secure access to sufficient amounts of safe and nutritious food for normal growth and development and an active and healthy life. It may be caused by the unavailability of food, insufficient purchasing power, inappropriate distribution or inadequate use of food at the household level. Food insecurity, poor conditions of health and sanitation and inappropriate care and feeding practices are the major causes of poor nutritional status. Food insecurity may be chronic, seasonal or transitory.”⁵⁶

Displaced persons

Displaced persons refer to the forced movement of people from their locality or environment and occupational activities. It is a form of social change caused by a number of factors, the most common being armed conflict. Natural disasters, famine, development and economic changes may also be a cause of displacement.⁵⁷

Undernutrition

Undernutrition is defined by the The United Nations Children's Fund (UNICEF) as the “outcome of insufficient food intake and repeated infectious diseases. Undernutrition includes being underweight for one's age, too short for one's age, dangerously thin, and deficient in vitamins and minerals. The term malnutrition refers to both undernutrition and overnutrition.”⁵⁸

History of Problem

Food insecurity has been an issue in our world since the beginning of time. Vast areas of the world have been plagued by an inequitable distribution of food sources thus led to food insecurity in countries that did not have access to safe and nutritious food. There are

⁵⁶ The Food and Agriculture Organisation “Glossary of selected terms”
<http://www.fao.org/hunger/glossary/en/> Accessed 6th December 2016

⁵⁷ UNESCO “Displaced Person / Displacement” <http://www.unesco.org/new/en/social-and-human-sciences/themes/international-migration/glossary/displaced-person-displacement/>
Accessed 23 December 2016

⁵⁸ The United Nations Children's Fund “Nutrition, Survival and Development”
https://www.unicef.org/progressforchildren/2006n4/index_undernutrition.html

a multitude of factors that contribute to food insecurity in our globalised world, such as climate change, military conflicts, cash crop dependence, corruption and political instability.⁵⁹

There are various regions which suffer from chronic food insecurity. One such example is The Sub-Saharan Africa (SSA) region, which has the highest prevalence of hunger - 1 in 4 people in this region are undernourished. Sub-Saharan Africa is the only region of the world where hunger is projected to worsen over the next two decades unless some drastic measures are taken to ensure peace, improve governance and achieve the economic development required to reverse the current trend⁶⁰.

This problem stems from a mix of different economical, political and social problems which eventually culminates in food insecurity, resulting in widespread hunger. Citizens in SSA have low average incomes but still rely on costly food imports which results in many citizens being unable to afford food. This is exacerbated by highly fragmented farming, lack of modern farming and productivity-enhancing techniques as well as weak transport infrastructure, logistics services and distribution channels⁶¹. All these factors contribute to making the SSA one of the most food-insecure region.

These factors are also applicable to various other food insecure regions, such as Asia, which has $\frac{2}{3}$ of its population hungry.⁶² There are 490 million people still suffering chronic hunger in the region, and the Asia-Pacific region is home to almost 62 percent of the world's undernourished. Ironically, the number of people who are overweight or obese is also rapidly increasing in the region, especially in Southwest Pacific Island countries and middle-income countries of Asia⁶³. Despite all these, the Asia-Pacific region has achieved the Millennium Development Goals' hunger target (MDG-1c) of halving the proportion of undernourished people by 2015. The region has also achieved the largest reduction in the absolute number of undernourished people (236 million)⁶⁴. This is the result of policy commitments, and institutional and technological innovations, combined with effective social protection measures.

Food insecurity can, reportedly, result in several devastating consequences, amongst which includes severe undernutrition and even displacement. While food insecurity has most usually called various agencies to attention for the alarming population numbers which are undernourished, it is truly the increasingly growing population of those who are displaced as a result of food insecurity which has called the international community to attention.

59 Harvest Help African Food Issue "Causes of Food Insecurity in African and Other Third World Countries" <http://www.harvesthelp.org.uk/causes-of-food-insecurity-in-african-and-other-third-world-countries.html> Accessed 13th December 2016

60 Food Security and Agricultural Development in Sub-Saharan Africa "Building a case for more public support Policy Brief No. 1" <https://www.oecd.org/tad/agricultural-policies/36784159.pdf> Accessed 13th December 2016

61 The Economist Intelligence Unit "Food security in focus: Sub-Saharan Africa 2014" [http://foodsecurityindex.eiu.com/Home/DownloadResource?fileName=EIU%20GFSI%202014 Sub-Saharan%20Africa%20regional%20report.pdf](http://foodsecurityindex.eiu.com/Home/DownloadResource?fileName=EIU%20GFSI%202014%20Sub-Saharan%20Africa%20regional%20report.pdf) Accessed 13th December 2016

62 World Food Programme "Hunger statistics" <https://www.wfp.org/hunger/stats> Accessed 13th December 2016

63 FAO "Regional Overview of Food Insecurity Asia and the Pacific" <http://www.fao.org/3/a-i4624e.pdf> Accessed 23 December 2016

⁶⁴ibid

Along with securing food, it is also important to note that water security is closely intertwined with our topic of discussion. This is especially important for agricultural areas, where water is a precious and important resource not only for personal survival but also for generating wealth and providing sustenance⁶⁵. In regions where water security cannot be achieved, food security is usually also affected and with it, the livelihood and survival of whole communities. While it is important to recognize the need for water security, the dais would like to remind delegates that the focus of the discussion should be on food security and the resultant displacement of peoples.

Statement and Discussion of Problem

Food insecurity is still a prominent issue in our globalised world. Even though progress continues in the fight against hunger, an unacceptably large number of people still lack the food they need for an active and healthy life. The latest available estimates indicate that about 795 million people in the world – just over one in nine people – were undernourished in 2014–2016.⁶⁶ Furthermore, more than 90 million children under the age of five are undernourished and underweight.

Undernourishment around the world, 1990–92 to 2014–16										
	Number of undernourished (millions) and prevalence (%) of undernourishment									
	1990–92		2000–02		2005–07		2010–12		2014–16*	
	No.	%	No.	%	No.	%	No.	%	No.	%
WORLD	1 010.6	18.6	929.6	14.9	942.3	14.3	820.7	11.8	794.6	10.9
DEVELOPED REGIONS	20.0	<5.0	21.2	<5.0	15.4	<5.0	15.7	<5.0	14.7	<5.0
DEVELOPING REGIONS	990.7	23.3	908.4	18.2	926.9	17.3	805.0	14.1	779.9	12.9
Africa	181.7	27.6	210.2	25.4	213.0	22.7	218.5	20.7	232.5	20.0
Northern Africa	6.0	<5.0	6.6	<5.0	7.0	<5.0	5.1	<5.0	4.3	<5.0
Sub-Saharan Africa	175.7	33.2	203.6	30.0	206.0	26.5	205.7	24.1	220.0	23.2
Eastern Africa	103.9	47.2	121.6	43.1	122.5	37.8	118.7	33.7	124.2	31.5
Middle Africa	24.2	33.5	42.4	44.2	47.7	43.0	53.0	41.5	58.9	41.3
Southern Africa	3.1	7.2	3.7	7.1	3.5	6.2	3.6	6.1	3.2	5.2
Western Africa	44.6	24.2	35.9	15.0	32.3	11.8	30.4	9.7	33.7	9.6
Asia	741.9	23.6	636.5	17.6	665.5	17.3	546.9	13.5	511.7	12.1
Caucasus and Central Asia	9.6	14.1	10.9	15.3	8.4	11.3	7.1	8.9	5.8	7.0
Eastern Asia	295.4	23.2	221.7	16.0	217.6	15.2	174.7	11.8	145.1	9.6
South-Eastern Asia	137.5	30.6	117.6	22.3	103.2	18.3	72.5	12.1	60.5	9.6
Southern Asia	291.2	23.9	272.3	18.5	319.1	20.1	274.2	16.1	281.4	15.7
Western Asia	8.2	6.4	14.0	8.6	17.2	9.3	18.4	8.8	18.9	8.4
Latin America and the Caribbean	66.1	14.7	60.4	11.4	47.1	8.4	38.3	6.4	34.3	5.5
Caribbean	8.1	27.0	8.2	24.4	8.3	23.5	7.3	19.8	7.5	19.8
Latin America	58.0	13.9	52.1	10.5	38.8	7.3	31.0	5.5	26.8	<5.0
Central America	12.6	10.7	11.8	8.3	11.6	7.6	11.3	6.9	11.4	6.6
South America	45.4	15.1	40.3	11.4	27.2	7.2	ns	<5.0	ns	<5.0
Oceania	1.0	15.7	1.3	16.5	1.3	15.4	1.3	13.5	1.4	14.2

*Data for 2014–16 refer to provisional estimates.
Source: FAO.

Fig.3: Changes in numbers and prevalence of undernourishment worldwide from 1990-2016
(Source: FAO⁶⁷)

⁶⁵ United Nations Water for Life Decade “Water and food security”
http://www.un.org/waterforlifedecade/food_security.shtml Accessed 13th December 2016

⁶⁶ The Food and Agricultural Organisation “The State of Food Insecurity in the World”
<http://www.fao.org/3/a-i4646e.pdf> Accessed 6th December 2016

⁶⁷ Ibid.

That said, there is, recognizably, a severe disparity between the information present and the actual amount of information collected. There are currently no internationally published statistics on the amount of displaced persons as a result of food insecurity. Without the availability of information, it is difficult for governments and international organisations to come up with a comprehensive framework to tackle displacement as a result of food insecurity.

Food insecurity has many devastating consequences especially in Less Economically Developed Nations (LEDs). Due to food insecurity, 66 million primary school-age children attend school hungry across the developing world – in Africa alone, 23 million children attend classes while hungry. As a result, the World Food Programme estimates that US\$3.2 billion each year is needed to reach all 66 million school-age children.⁶⁸

Across all countries the percentage of undernourished people has dropped over the years but it is still not at a satisfactory level. Currently the percentage of undernourished people is at 10.9% globally, far above the 5% deemed as “very low” by the Millennium Development Goal hunger target.⁶⁹ More measures and policies must be implemented in order to ensure that more countries and people are food secure.

Delegates must take note that national self-sufficiency is neither necessary nor sufficient to guarantee food security at the individual level. For example, countries like Hong Kong and Singapore are not self-sufficient (agriculture is nonexistent) but their populations are food-secure, whereas India is self-sufficient but a large part of its population is not food-secure.⁷⁰ As a result, about 21% of India's population are suffering from undernutrition⁷¹; an outcome of long term food insecurity as they continuously lacked access to sufficient amounts of safe and nutritious food. This thus shows that a country's agricultural assets do not often directly relate to its food security and displacement rates, and instead depends on a multitude of intertwined economical, political and social problems which remains to be resolved.

Current Situation

The World Food Programme (WFP) has characterized hunger, or food insecurity, to stem from 6 key factors; the poverty trap, lack of investment in agriculture, climate and weather, war and displacement, unstable markets and food wastage. Each of these factors clearly affect at least one of the four key aspects of food security: availability, access,

68 World Food Programme “Hunger Statistics” <https://www.wfp.org/hunger/stats>
Accessed 13th December 2016

69 Food and Agricultural Organisation “The State of Food Insecurity in the World 2015”
<http://www.fao.org/hunger/en/> Accessed 6th December 2016

70 Global food security under climate change <http://www.pnas.org/content/104/50/19703.full>
Accessed 12th December 2016

71 William Thomson “India's Food Security Problem”
<http://thediplomat.com/2012/04/indias-food-security-problem/> Accessed 12th December 2016

stability and utilization in some way⁷². Recently however, much focus on dealing with food security has been poured into the study of climate change and its effects on food security.

Climate change poses serious threats to agriculture and food security globally. Its impacts on agriculture include, but are not limited to, heat waves, pests, drought, desertification, freshwater decline, and biodiversity loss. The global poor, who are most dependent on agriculture for their livelihoods, are most vulnerable to climate change impacts on agriculture. They are also the most likely to be forced from their homes when a drought or flooding wipe out agricultural resources on which they depend, effectively becoming food insecure and displaced⁷³.

A recent study concluded that if climate change continued on the path it is on, its impact on food security would lead to approximately 529,000 deaths by 2050⁷⁴. While there are no reliable global estimates for the number of people uprooted each year due to food insecurity brought on by drought, severe flooding, or other climate-related disasters, there are numerous tragic examples of the propensity of climate change-related effects to displace significant numbers of poor and vulnerable people.

Examples of these include the 2011 drought and famine in the eastern Horn of Africa, killing an estimated 260,000 Somalis and driving a quarter of the country's population from their homes. Even today, and despite the fact the Somali government has now gained control of large parts of the country, one in ten Somalis remains displaced either within Somalia or abroad, many of whom are food insecure⁷⁵. In 2014-2015, the Andhra Pradesh and Tamil Nadu floods in India displaced over 1.8mil people in the Tamil Nadu state⁷⁶, with millions of hectares of crops destroyed, exacerbating the food security situation in India.

Climate-induced migration and displacement surfaced as a key concern at the 2015 United Nations Climate Change Conference (known as COP21) that took place in early December. In his opening remarks, President Obama noted the possible futures we face if climate change remains unaddressed: "Submerged countries. Abandoned cities. Fields that no longer grow. Political disruptions that trigger new conflict, and even more floods of desperate peoples seeking the sanctuary of nations not their own."⁷⁷ Some reports even suggest that climate change could cause the creation of a new type of refugee, appropriately named "climate refugee", of which populations in low-lying coastal regions, such as in Bangladesh or the Maldives, are most likely to become a part of.

⁷² World Food Programme. (2016). *What causes hunger*. Retrieved 12 12, 2016, from [https:// www.wfp.org/hunger/causes](https://www.wfp.org/hunger/causes)

⁷³ Pulaski, T. (2016, March 8). *Food Security and Displacement in a Warming World*. Retrieved October 30, 2016, from <http://www.refugeesinternational.org/blog/2016/3/7/foodsecurity>

⁷⁴ Springman, M. (2016). *Global and regional health effects of future food production under climate change: a modelling study*. The Lancet.

⁷⁵ Pulaski, T. (2016, March 8). *Food Security and Displacement in a Warming World*. Retrieved October 30, 2016

⁷⁶ IDMC. (2015). *Disaster Induced Displacement Dataset. IDMC's Global Internal Displacement Database*. IDMC.

⁷⁷ Heimann, L. (2015, December 10). *Natural Disasters Displace Millions, Affect Migration Flows*. Retrieved October 30, 2016, from <http://www.migrationpolicy.org/article/top-10-2015-issue-7-climate-change-and-natural-disasters-displace-millions-affect>

Widely cited estimates show that over the period 1980 to 2008 there was a 5.5 percent drop in wheat yields and a 3.8 percent drop in maize yields globally, compared to what they would have been had climate remained stable⁷⁸. Additionally, the IPCC Fifth Assessment Report provides new evidence that crop yields are expected to decline in areas that already suffer food insecurity. Climate change also affects livestock production in multiple ways, of which the most important impacts are on animal productivity, animal health and biodiversity, the quality and amount of feed supply, and the carrying capacity of pastures.

Having understood the threats that climate change can pose to both food security and mass displacement of human populations, the UN Paris Climate Change Agreement was adopted by national governments. Even though the Paris Agreement did see nations commit to cutting greenhouse gas emissions to prevent a 2-degree Celsius increase in global temperatures, it is clear frameworks that deal with food security and climate-induced displacement require much more international support.

Case Studies

For India, food security continues to be high on its list of development priorities because the country's relatively high rates of economic growth have not led to a reduction in hunger and undernutrition. India's gross domestic product at factor cost and per capita income grew at seven percent and five percent per annum, respectively, from 1990-91 to 2013-14. However, the incidence of undernutrition has dropped only marginally from 210.1 million in 1990 to 194.6 million in 2014, and India has failed to meet the Millennium Development Goal of halving the proportion of people who suffer from hunger. About 12 Indian states fall under the 'alarming' category of the Global Hunger Index. According to the National Family Health Survey 2015-16, the proportion of children under five years who are underweight is significantly high in states such as Bihar (43.9 percent), Madhya Pradesh (42.8 percent) and Andhra Pradesh (31.9 percent)⁷⁹.

Even so, the FAO has rated the overall food security situation in India is to be satisfactory. This is mainly due to ample food stocks and the provision of highly-subsidized rice, wheat and coarse grains through distribution programmes, especially for the below-poverty line families through the National Food Security Act (NFSA). Still, it acknowledges that India is prone to floods during the monsoon season, which normally affect large numbers of people. In recent years, recurrent floods and landslides mostly concentrated in the northeastern parts of the country, affected at least 5.9 million people in Assam and Bihar alone, causing deaths and injuries, mass displacement (1.8mil), loss of livestock and flood supplies⁸⁰.

Past UN Actions

The UN FAO acts as the main UN body dealing with issues relating to food security, while the UNHCR deals with climate-induced internally displaced persons (IDPs). Both

⁷⁸Lobell, D., Schlenker, W., & Costa-Roberts, J. (2011). *Climate Trends and Global Crop Production since 1980*. PubMed.

⁷⁹Chakrabarty, M. (2016). *Climate Change and Food Security in India*. ORF.

⁸⁰ FAO. (2016, August 24). *GIEWS Country Briefs on India*. Retrieved October 30, 2016, from <http://www.fao.org/giews/countrybrief/country.jsp?code=IND&lang=en>

organisations have a variety of different programmes and framework aimed towards dealing with the issues of concern, often tailored to a particular region or nation.

FAO Mitigation of Climate Change in Agriculture (MICCA) Programme

The MICCA programme strengthens FAO's longstanding work to address climate change in the agriculture, forestry and fisheries sectors and supports countries participating in the climate change negotiation processes within the United Nations Framework Convention on Climate Change⁸¹.

The MICCA programme generates technical knowledge, working on the ground and with partners to:

1. Monitor and assess greenhouse gas (GHG) emissions and the mitigation potential in agriculture;
2. Develop the capacity of stakeholders working on national GHG inventories and farmers using Climate-Smart Agriculture (CSA) practices;
3. Carry out life cycle assessments to guide decision making;
4. Give guidance on climate change mitigation & adaptation options, including for peatlands and organic soils;
5. Mainstream gender in CSA; facilitate online communities of practice; and run online learning events.

MICCA implemented two pilot projects in the United Republic of Tanzania and Kenya in 2011-2014 providing evidence that climate-smart agricultural practices can reduce GHG emissions, improve farmers' lives and make local communities better able to adapt to changing climatic conditions.

In Kenya, the four-year pilot project involved smallholder dairy producers in Kaptumo in the Nandi South District of the Rift Valley Province. The pilot project, which interacted with over 4 500 male and female farmers, was carried out within the framework of the East Africa Dairy Development Project (EADD).

In this pilot project, the MICCA Programme, ICRAF and smallholder dairy producers used a life cycle assessment, EX-ACT tool and other approaches to test technical alternatives for reducing the climate change 'footprint' of the dairy industry.

As part of its work to support farmers in raising climate-smart cattle and to improve the overall greenhouse gas balance of the farming systems, the pilot project was looking at ways to enhance fodder production and livestock management and to improve the management of manure to produce fertilizers and biogas. The project resulted in reduced GHG emissions from the farm systems and enhanced sinks⁸².

MICCA played a significant role in enriching the general body of knowledge on key GHG sources in the Agriculture, Forestry and Other Land Use (AFOLU) sector, and enables a more comprehensive understanding of the mechanisms that cause emissions. Additionally, farmers who participated in the MICCA pilot projects reported that the main benefits of CSA were higher yields, greater farm income and increased food availability. These results

81 FAO. (2016). *Mitigation of Climate Change in Agriculture (MICCA) Programme*. Retrieved October 30, 2016, from <http://www.fao.org/in-action/micca/en/>

82 FAO. (2016). *Kenya | FAO | Food and Agriculture Organization of the United Nations*. Retrieved October 30, 2016, from <http://www.fao.org/in-action/micca/on-the-ground/africa/kenya/en/>

indicate that CSA can be an effective pathway for improving food security, alleviating poverty and building more resilient livelihoods.

Perhaps the main criticism that could be levelled against MICCA would be the current small take-up rate of the programme globally, with less than 10 countries implementing the suggestions made by MICCA into its national agricultural development policy. This demonstrates that without proper whole-of-government support for genuine investment in the agricultural sector, true food security still remains elusive for much of the rural world. Furthermore, while MICCA does well in targeting the pillars of food availability and stability in ensuring food security, it fails to address the pillars of proper access and utilization of food which collectively also make-up food security.

It is perhaps key to note that food security is not an exclusively rural problem and developed countries also have much to do in gradually improving global food security. In particular, most of the developed world have not done much in ensuring that food wastage, which accounts for one-third of all food produced and is a key contributor to global food insecurity, is significantly reduced.

Bloc Positions

Despite having much attention paid to developing countries, developed countries have also faced problems in achieving food security. In dealing with this topic, therefore, it is important to note the diverse nature of different nations, in relation their production, acquisition or sale of food products.

Europe and North America

Both regions are performing well in terms of ensuring food security. Notably, Europe has presented a set of comprehensive safety-net programmes to ensure food security within the region, and to ensure that food price inflation is not too drastic⁸³. Both Europe and North America has been seen to rank first and second of all regions in the world in terms of achieving adequate and optimal availability, as well as affordability.

That said, both regions are facing the problem of a highly volatile production and low urban absorption capacity. Urban absorption capacity is defined as the ability of a country to absorb urbanizing populations and to support their food needs.

These two regions also tend to be large contributors to food aid internationally. Significant amounts of food aid received in the Sub-Saharan African region tend to be contributed by this bloc, as these nations often possess both agricultural strength and also financial strength to offer such assistance.

Asia, Middle East, Central and South America, North Africa

These regions comprise of a mixture of developed and developing countries, each varying in terms if economic and political structures. In these regions, income levels are closely tied to food security. There can thus be observed the trend that the more developed a country is, the better they fare in terms of food security.

⁸³The Economist. (2014). *Food Security in Focus: Europe 2014* (Rep.). Retrieved November 2, 2016

That said, corruption is a problem that greatly affects food distribution, and significantly distorts the price of the food products – this is seen especially in the regions of Central and South America. Despite so, however, the regions have performed well in dealing with price and supply volatility, despite failures in crop harvests.

Several Middle Eastern and North African countries are also performing well in terms of ensuring food security. States such as Turkey and Morocco are self-sufficient in almost all agricultural products. That said, Gulf Nations remain heavily reliant on food imports, but this reliance is tempered with reduced agricultural tariffs to ensure affordability.

Sub-Saharan Africa

This region has been reported to constantly face an inability to meet the factors of: affordability, availability, as well as quality and safety. Part of this is due to the large percentage of low-income countries within the region – it is reported that of the 28 countries within the region, 18 of them are considered to be of low-income and are unable to ensure constant ability to afford for food products.

The main problem faced by SSA is a widespread poverty and reliance on food imports that are often too costly for the locals. Moreover, populations in the SSA sprawl over large areas which, more often than not, face weak infrastructures which make food distribution exceedingly difficult. Political and social unrest is also rampant within this region, and frequently affect food distribution. This has resulted in a surge in the number of refugees and internally displaced people.

That said, SSA has also experienced a rapid and significant economic growth in the recent years, and this has resulted in an overall strengthening of food security in the region. It should also be noted that the SSA has an abundance of natural resources, and local authorities often thus seem to focus their policies on the agricultural sector so as to increase affordability and availability of food products.

Overall, this region is the largest recipient of food aid.

Questions a Resolution Must Answer

1. Can current levels of production meet future food needs?
2. Is national food security still critical given global trade? Are agricultural agreements still relevant?
3. Will relying on external trade for food affect food security in a nation by reducing employment?
4. Does free trade of food products benefit those most likely to suffer from food insecurity?
5. What is the role of food aid, given the increasing prominence of trade? (Note: aid is increasingly being thought of as distorting markets of recipient states. NGOs have, very often, pushed for a “trade, not aid” approach. However, trade subjects nations and/or regions to volatility and uncertainty – are there ways to go about solving this problem? How then to ensure that trade will not only facilitate the continued dominance of oligopolies?)